STAT 359 Final Project

FinBERT Under Geopolitical Domain Shift: Evaluating LoRA and Fine-Tuning for Cross-Domain Sentiment

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Project Overview

Project Overview

Project Track:
Engineer Track
(Mechanistic Focus)

Model:

FinBERT
(ProsusAI/finbert)

Aims and Motivations of this Project:

This project tests if fine-tuning FinBERT for financial sentiment improves its ability to interpret geopolitical news headlines.

Specifically performance evaluations of increased adaptation capacity in FinBERT via Lora and full fine tuning.

Overview of Methodology

Methodology

Training data collection

Financial PhraseBank from HuggingFace (takala/financial_phrasebank, sentences_allagree split)

Composed of approx 2.2k sentences labeled by annotators into 3 classes: positive, negative, neutral.

A standard 80/20 train/validation split, seed=42 was implemented

Test data collection

Data Collection

Created article collection with filters for specific time periods, and themes, including:

- COVID, Brexit and Trade War 2018–2021
- Semiconductor controls, Ukraine sanctions 2022–2023
- Trump tariffs and hostile foreign policy 2024–2026

Data Processing

Dropped irrelevant articles via defined rubric for labeling.

Characteristics:

- Sample size = 178 articles post-processing
- Label distribution: Neg 57%, Neu 25%, Pos 17%

News article dataset deliberately not used in training - used for model evaluation

Methodology

LoRA Rank & Epoch

LoRA r=4: low-capacity adapter baseline

LoRA r=16: higher-capacity parameter-efficient adaptation

Full fine-tuning: all parameters updated baseline

5 epochs
Batch size 16
Learning rate 2e-5

Initial Training on Financial PhraseBank

Trained all 4 models on Phrasebank to ensure:

- in-domain performance is sound
- Provides baseline for Out of Domain testing

Model Results and Performance

Zero Shot Baseline Performance

FinBERT without any additional training evaluated directly on 178 geopolitical headlines.

Performance published benchmarks on standard financial text (Guo et al. report macro F1=0.555).

Class	F1	Support
Pos	0.36	31
Neu	0.45	45
Neg	0.62	102
Macro F1	**0.49**	178

Key Findings

Negativity Bias

Baseline identifies negative events reasonably well (Negative F1 = 0.62). Due to similarities in financial and geopolitical language.

Positive Event Fails

Fails on positive resolution events (Pos F1=0.36). De-escalation language e.g. trade deal pauses, diplomatic progress out of FinBERT's pre-training distribution.

Weak Neutral Classification Performance

Neu performance (0.45) is weak.

LoRA F1-Scores Analysis

Performance/Scores

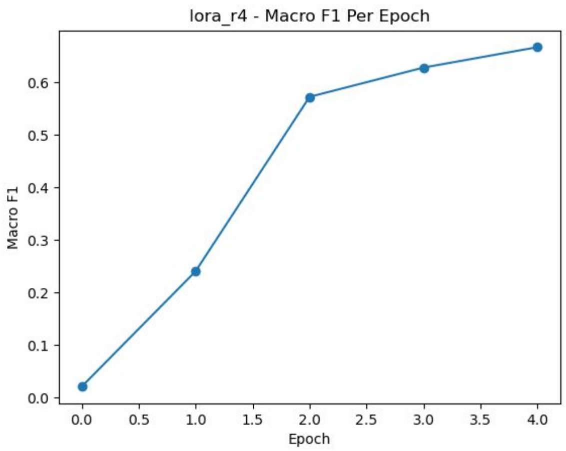
- LoRA r4 Val Macro-F1 ≈ **0.66**
- LoRA r16 Val Macro-F1 ≈ **0.89**

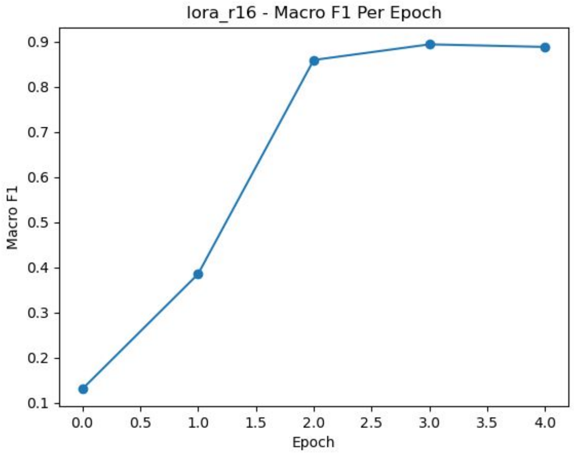
Key Findings

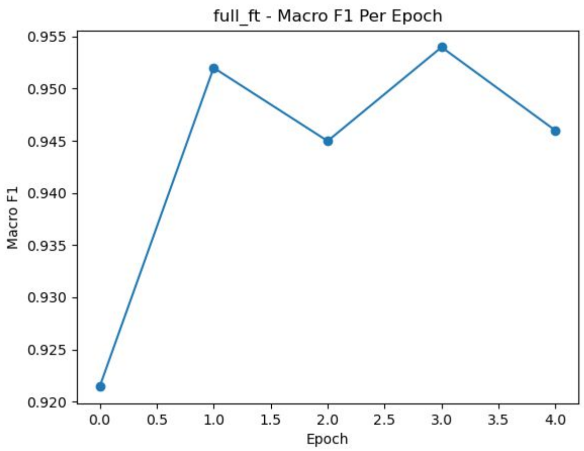
Based off these scores, findings include…

Increasing LoRA rank improves in-domain sentiment classification performance on Financial PhraseBank.

Higher rank allows the model to learn richer task-specific adaptations.







LoRA Confusion Matrix Analysis

Performance/Scores

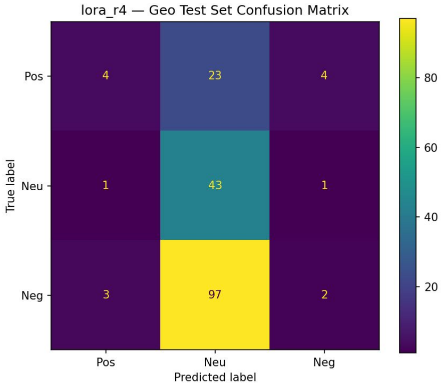
- LoRA r4 Geo Confusion True Neg: predicts Neutral 97 times
- LoRA r16 Geo Confusion True Neg: predicts Neutral 94 times
- Both models predict Neutral for nearly all geopolitical headlines.

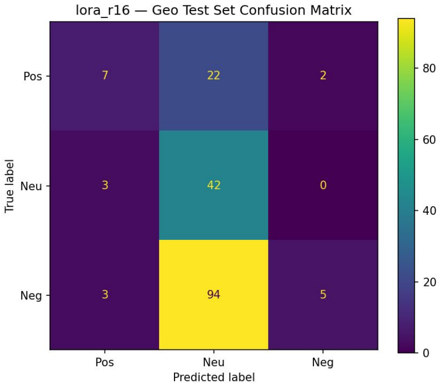
Key Findings

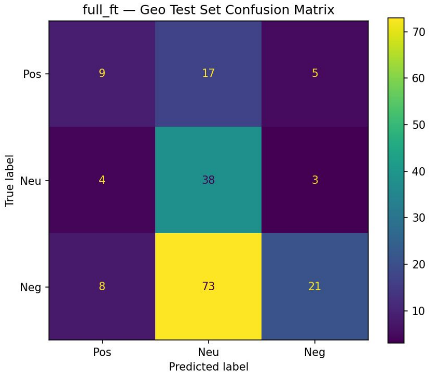
Based off these results, findings include…

Under geopolitical domain shift, both LoRA models strongly default to the Neutral class, due to ambiguity.

Increasing LoRA rank helps fit the training domain but does not substantially improve transfer to geopolitical headlines.







Live Demo

Example 1: Negative geopolitical event
Headline: “U.S. expected to impose up to $60 billion in China tariffs by Friday: sources”

Model	Prediction	Confidence
LoRA r=4	Neutral	0.85
LoRA r=16	Neutral	0.91
Full Fine-Tuned	Neutral	0.997

Example 2: Positive geopolitical event
Headline: “Trump claims ‘total reset’ of US-China ties as 90-day pause to trade war agreed”

Model	Prediction	Confidence
LoRA r=4	Neutral	0.84
LoRA r=16	Neutral	0.91
Full Fine-Tuned	Neutral	0.996

LoRA Training Curves Analysis

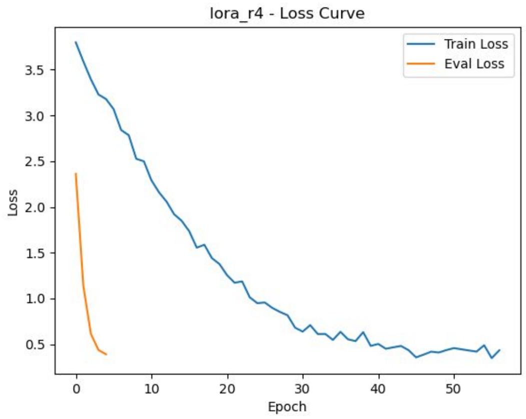
Performance/Scores

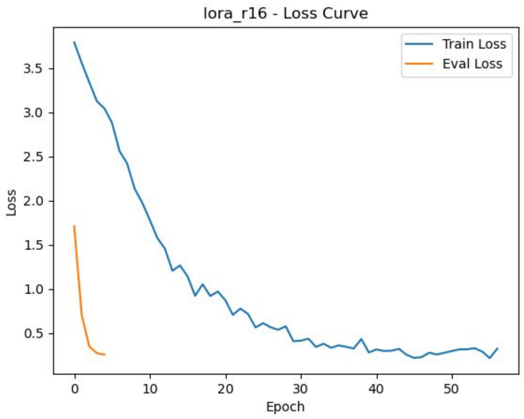
- All three training loss curves witness steady decrease, minimal oscillation or divergence
- Sharp drops in all three eval loss curves, then stabilizes

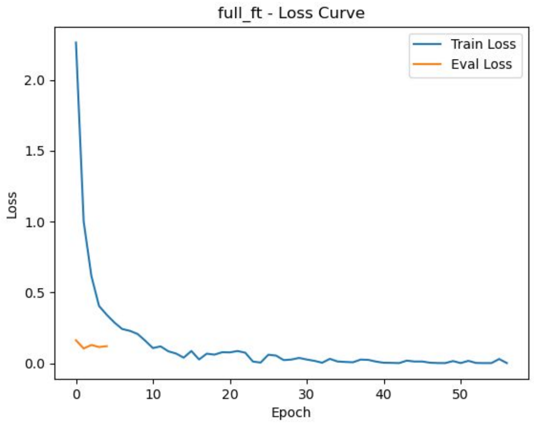
Key Findings

Based off these scores, findings include...

- All three configurations successfully learn the Financial PhraseBank task, as indicated by steadily decreasing training loss
- Indicates models converge quickly on the Financial PhraseBank task.
- Full fine-tuning have all parameters updated
- Stronger domain fit does not mean better geopol classification







Insights, Limitations and Conclusion

Insights & Limitations

Key Insights

A strong in-domain performance does not transfer to performance on geopolitical headlines

However, increasing adaptation capacity improves in-domain performance.

Full fine-tuning partially improves transfer, less neutral classification but does not solve the problem

Limitations

Sample size and distribution: 178 headlines, small positive class size - larger dataset may help provide more robust evaluation

Data in Financial Phrasebank and Geopolitical Headlines dataset have different characteristics.

LoRA geo results artefactual due to adapter loading failure - cannot currently compare LoRA vs full_ft on domain shift

Conclusion

• Increasing adaptation capacity (LoRA r=4 to r=16 to full fine-tuning) significantly improves in-domain financial sentiment performance.

• Performance does not transfer well to geopolitical headlines - highlights clear domain shift between financial and geopolitical language.

• LoRA models exhibit parameter-efficient adaptation. Nonetheless, higher ranks do not substantially improve geopolitical classification.

• All models show a strong bias toward predicting neutral sentiment - suggests geopolitical sentiment is more nuanced than financial sentiment.

• Results suggest that for a more robust analysis, a domain-specific training data or domain-adaptive fine-tuning is needed

References

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LoRA: Low-Rank Adaptation of Large Language Models.

Thank you!

Q&A